

Paper Reading Template — EHR-Based Pre-Consultation Medical Documentation System (Bangla)

Instructions for the AI Reading the Paper

You are helping a **beginner ML/NLP team** build a Bangla speech-to-EHR system. Project context:

- **Pipeline:** Patient speaks → ASR → Medical NER → Structured EHR → (optional) differential diagnosis hints for the doctor
- **Languages:** Bangla (primary) + Bangla-English code-mixed
- **Dialects:** Dhaka, Sylhet, Chittagong, Barishal, etc.
- **Resources:** Zero budget, Google Colab only (T4 GPU, ~15GB VRAM, ~12h session limit)
- **Data:** ~200-300 self-collected Bangla medical audio recordings
- **Entities to extract:** symptoms, diseases, medications, duration, allergies

Rules for filling this form

1. If a field is **not applicable** to the paper → write N/A.
2. If you are **not sure** / **the paper is ambiguous** → write "**Unsure: [reason]**". Do NOT guess silently.
3. If an entire section is irrelevant → write "**Not covered in paper**" and skip subfields.
4. **Be brief.** Bullet points > paragraphs. **Paraphrase**, do not copy text from the paper.
5. Always end each major section with a one-line "**Why this matters for our project.**"
6. At the very end, list every place you were unsure (Section 13).

1. Basic Information

- **Title:**
 - **Authors:**
 - **Affiliation / Venue / Journal:**
 - **Year:**
 - **Paper link:**
 - **Code / model link (if any):**
-

2. Quick Relevance Snapshot

Primary domain(s) covered (tick all that apply):

- ☐ Speech Recognition (ASR)
- ☐ Medical NLP / Clinical Information Extraction
- ☐ Bangla / Low-Resource NLP
- ☐ Accent / Dialect handling
- ☐ Code-Switching / Code-Mixing
- ☐ EHR / Clinical Documentation
- ☐ Differential Diagnosis / Clinical Reasoning
- ☐ Other: _____

Relevance scores (1 = irrelevant, 5 = directly usable in our project):

Aspect	Score	One-line reason
ASR for our pipeline	/5	
Medical entity extraction	/5	
Bangla / low-resource techniques	/5	
Dialect / code-mixing	/5	
EHR structuring	/5	
Feasible on free Colab	/5	

Overall verdict: Deep-study / Partial-read / Skim / Skip — *why*:

3. Problem & Solution

A. What problem does the paper solve? (2–3 sentences, plain English)

B. Their core approach in 2–4 sentences:

C. Connection to our project:

- Direct application:
 - Technique we can borrow:
 - Where their setting differs from ours:
-

4. Methodology

Fill ONLY the subsections (4A-4D) that the paper actually covers. Mark the others "Not covered."

4A. Speech Recognition (*only if the paper does ASR*)

- ASR model (e.g., Whisper, Wav2Vec2, Conformer, MMS):
- Pre-trained on what data?
- Fine-tuned on what data?
- Language model / decoder used:
- Data augmentation (SpecAugment, noise, speed perturb, etc.):
- How they handle **accents** / **dialects**:
- How they handle **code-switching**:

4B. Medical NLP / NER (*only if the paper does medical IE*)

- Model (BiLSTM-CRF, BERT, mBERT, BanglaBERT, LLM, rule-based, hybrid):
- Entity types extracted (symptoms, meds, dose, duration, allergies, diagnosis...):
- Annotation scheme (BIO / BIOES / span / other):
- Medical terminology source (UMLS / SNOMED / ICD / custom dictionary / learned):
- Output format (JSON / FHIR / template / other):

4C. Low-Resource / Cross-Lingual / Code-Mixing tricks (*only if relevant*)

- Data-scarcity solutions (transfer learning, augmentation, synthetic data, back-translation, self-training):
- Cross-lingual transfer (source → target language):
- Code-mix handling:

4D. Differential Diagnosis / Clinical Reasoning (*only if covered*)

- Method (rule-based, knowledge graph, ML classifier, LLM, retrieval):
- Knowledge source (ICD, textbooks, expert rules, curated dataset):

4E. Pipeline diagram (in text)

Input → [Step 1] → [Step 2] → ... → Output

Why this matters for our project:

5. Data

- **Dataset name:**
- **Size** (hours of audio / number of samples):
- **Language(s):**
- **Domain:** general / medical / other
- **Dialect / accent coverage:**
- **Publicly available?** Yes / No / Restricted — link:
- **Collection method:** clinical / crowdsourced / simulated / scraped
- **Annotation process** (who annotated, inter-annotator agreement, guidelines):
- **Preprocessing steps:**

Can we use this dataset? Yes / No / Partial — *why*:

6. Evaluation

Only fill metrics actually reported in the paper. Others = N/A.

Category	Metric	Score
ASR	WER	
ASR	CER	
NER / IE	Precision	
NER / IE	Recall	
NER / IE	F1	
NER / IE	Entity-level F1	
Clinical / EHR	clinical accuracy / doctor satisfaction / time saved	
Other		

- **Train / Val / Test split:**
 - **Hardware used:**
 - **Framework** (PyTorch, HuggingFace, ESPnet, Kaldi, other):
-

7. Key Results & Findings

- **Best result achieved:**
 - **Compared to which baselines?** By how much better?
 - **What worked well:**
 - **What didn't work / failure cases:**
 - **Error analysis highlights:**
 - **Authors' stated limitations:**
-

8. Reproducibility & Feasibility for Us

- **Code available?** Yes / No / Partial — link:
 - **Pre-trained models available?** Yes / No — link:
 - **Rough GPU need:**
 - **Will this run on free Colab (T4, ~15GB VRAM, 12h)?** Yes / Maybe / No — *why*:
 - **Data accessible to us?** Yes / No / Partial:
 - **Estimated time to reproduce a small/demo version:** hours / days / weeks
 - **Feasibility verdict:** Fully reproducible / Partial / Demo only / Not feasible
-

9. Direct Takeaways for Our Project

Things we can copy / adapt directly: 1. 2. 3.

Things to avoid / not applicable to us: 1. 2.

Open questions this paper raises for us: 1. 2.

Which phase of our project does this help? (tick all)

- ☐ ASR baseline setup (Whisper / Wav2Vec2 / etc.)
 - ☐ Bangla ASR fine-tuning
 - ☐ Dialect / code-mix handling
 - ☐ Data collection & annotation strategy
 - ☐ Medical NER
 - ☐ EHR structuring (JSON / template)
 - ☐ Differential diagnosis hints
 - ☐ Evaluation methodology
 - ☐ System integration
-

10. Action Items

Concrete next steps after reading this paper:

- 1.
 - 2.
 - 3.
-

11. Questions for Supervisor / Team

For supervisor (AZK):

-

For team discussion:

-
-

12. Keywords & Concepts

Keywords from the paper relevant to our project (≥ 5): 1. 2. 3. 4. 5.

Technical terms our team needs to learn: 1. 2. 3.

13. AI Reader's Confidence Note (REQUIRED)

List every place you were unsure, guessed, or where the paper was ambiguous. Be honest — this helps the team know which fields to double-check.

- Section ___ : unsure because ___
- Section ___ : paper didn't clearly state ___
- Section ___ : my guess based on context, not stated explicitly ___

(If everything was clear, write: "All sections answered with high confidence.")